AMAN TIWARI

Geospatial Professional | AI / ML | Google Earth Engine | Python

+91-8859962467 | amantiwari.contact@gmail.com | LinkedIn | Delhi, India

SUMMARY

Remote Sensing & GIS specialist with 4+ years building climate, agriculture and land-use analytics from multi-temporal, multi-sensor satellite imagery using Google Earth Engine (GEE), Python and PostGIS. Experienced in raster and vector data processing, vegetation analysis (NDVI, SAVI, MSAVI), land-cover (LULC) classification, change detection and time-series analysis. Delivered heat-stress mapping that informed a municipal Heat Action Plan, worked on building the next-gen AI enabled Highway monitoring system using drone analytics and building a carbon dMRV system from ground up by combining quantitative spatial modelling, reproducible data pipelines, QA/QC rigor and dashboard delivery with a record of presenting results to senior decision makers and collaborating with Product and Engineering teams.

EXPERIENCE

Product & GIS Manager | FCF India, New Delhi **Feb 2025 – Present**

- Lead satellite imagery, remote sensing and geospatial analytics delivery for a Digital MRV (dMRV) platform supporting VCS/VM0047 ARR and Nature-based Solutions (NbS) carbon projects, integrating satellite imagery, drone data and ground-truth field surveys across multiple Indian states.
- Built Python and Google Earth Engine pipelines for baseline and periodic monitoring: time-series analysis of NDVI, SAVI and MSAVI vegetation indices, LULC classification and multi-temporal change detection, used for parcel-level performance assessment and Verra Performance Benchmark calculations.
- Standardized GPS, plot and survey data integration (raster and vector datasets) with QA/QC protocols; produced maps, spatial evidence and technical documentation supporting validation/verification and audit readiness.
- Architected a scalable, cloud-based geospatial processing pipeline on GCP (Cloud Run, GCS, PostGIS) serving drone-derived imagery, cutting load time by 80%+; collaborated with Product and Engineering stakeholders; led and mentored a 3-analyst GIS team.

GIS Engineer | DronaMaps, Gurugram **Apr 2024 – Feb 2025**

- Led a Drone Analytics Monitoring System (DAMS) for NHAI, enabling multi-temporal, GIS-based change detection and monitoring across 11,000+ km of national highways.
- Deployed interactive geospatial web dashboards (ESRI Web App Builder) used by 100+ decision-makers; managed a 35-person team, raising data-processing throughput by 40%.

GIS Consultant | CEEW (Council on Energy, Environment and Water), New Delhi **Sep 2023 – Apr 2024**

- Conducted spatial and time-series analysis in Google Earth Engine and Python to identify high-heat-stress zones for Thane's Heat Action Plan, directly informing municipal policy.
- Designed the Climate Resilience Atlas for India, integrating 15+ geospatial data layers (raster and vector) to support climate risk analytics and adaptation strategies across 700+ districts.

GIS Executive | Attentive.ai, Remote **Mar 2021 – Sep 2021**

- Produced and QA'd GIS feature datasets from high-resolution satellite and aerial imagery, supporting AI-based automated mapping and land-feature extraction workflows.

GIS Consultant & Trainer | AGSRT, Remote **Sep 2020 – Mar 2021**

- Delivered GIS and remote-sensing training (QGIS, ArcGIS) to students and professionals and consulted on spatial analysis and survey-data integration workflows.

GIS Data Intern | Intensel, Remote **Jun 2020**

- Prepared and processed geospatial datasets supporting satellite-based climate and flood-risk analytics.

EDUCATION

M.Sc. Geo-Information Science & Earth Observation - University of Twente (ITC), Netherlands 2021 – 2023

B.Tech. Geoscience Engineering - University of Petroleum & Energy Studies (UPES), Dehradun 2016 – 2020

TECHNICAL SKILLS

Remote Sensing & GIS: Google Earth Engine (Python API & JavaScript), Satellite Image Processing, Remote Sensing, Raster & Vector Data, QGIS, ArcGIS Pro, GeoServer, GDAL, multi-temporal & multi-sensor optical analysis, change detection, Geo AI, LULC classification, time-series and trend analysis, vegetation indices (NDVI, SAVI, MSAVI), biomass estimation and modelling, flood mapping using SAR data.

Programming & Data: Python, Pandas, NumPy, GeoPandas, Rasterio, SQL, PostGIS, Spatial Databases, Data Visualization, JavaScript, Node.js, React, Streamlit

Climate & Risk Analytics: heat-stress and climate-vulnerability mapping, multi-layer hazard/risk indexing, flood-risk analytics, environmental monitoring, spatial QA/QC, risk dashboards and audit-ready maps

Agriculture & Carbon: Agriculture Analytics, Land-Use Analysis, ARR & AFOLU, VCS/Verra VM0047, MRV/dMRV workflows, ground-truthing, GPS-based field data collection, validation/verification support

Cloud & Tools: GCP (Cloud Run, GCS), AWS, Cloud-based Geospatial Processing, Cesium.js

Languages: English, Hindi

SELECTED PROJECTS

- 3D Flood Visualisation Web GIS (M.Sc. Thesis) - Web GIS framework for real-time 3D flood visualisation integrating satellite/terrain data for urban resilience and municipal flood-risk planning (Python, Cesium, Three.js).
- Thane Heat Action Plan – Published in CEEW.